

Harsh Verma (he/him)

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SUMMARY

Electrical Engineering undergraduate at NIT Hamirpur and an active GNOME contributor, currently building unit testing frameworks for GIMP and improving its CI pipelines. Works across Rust, Python, C, C++, and Kotlin, on everything from STM32 firmware tooling to full stack Android apps. Enjoys building developer tools and embedded software that stay simple to test, version, and automate.

EDUCATION

- **National Institute Of Technology** Hamirpur, India
Bachelor of Technology - Electrical Engineering *July '24 - May '28 (Expected)*
- **Vidya Bhawan Public School** Udaipur, India
Class 12th - Maths, Physics, Chemistry *May '22 - April '23*

TECHNICAL SKILLS

- **Languages:** Rust, Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Bash
- **Embedded & Hardware:** Embedded Systems, STM32 / ARM Cortex-M, PCB Design, Power Management Circuits, Raspberry Pi, arduino-cli, OpenOCD
- **Developer Tooling:** CLI Design, LSP / Language Servers, CMake, Unit Testing, Google Blockly, PyQt6, GIMP Toolkit (GTK+)
- **Backend & Databases:** Django, Firebase (Auth, Realtime DB, Firestore), MySQL, REST APIs, WebSocket
- **Frontend & Mobile:** Jetpack Compose, Navigation, React
- **Automation & DevOps:** GitHub Actions, GitLab CI/CD, GitHub API, Git & GitHub, Linux, FFmpeg

OPEN SOURCE EXPERIENCE

- **GNU Network Object Model Environment - GNOME** [GitLab \(@harshverma\)](#)
Active Contributor *February '26 - Present*
 - **Unit Testing - GIMP:** Built a *Unit Testing Framework* from scratch for core GIMP libraries, *covering 65+ files* and significantly improving test coverage; designed test cases to ensure functionality and edge cases, integrated the suite into *GitLab CI/CD* pipelines for *automated validation* on every commit, and identified a previously undetected environment-variable bug through the test analysis.
 - **Unit Testing - BABL & GEGL:** Discovered a crucial flaw in GitLab CI that allowed a single commit to spawn two parallel pipelines, and authored the fix, *reducing CI resource usage by 50%*.
 - **Blur Strength - GIMP:** Replaced traditional weighted blur algorithms with *Gaussian-distributed kernels* and refined kernel specifications to align better with user requirements.

PROJECTS

- **Nucleus** – *A modern STM32 developer platform (In Progress)*: Building a CLI-first, version-controllable alternative to STM32CubeIDE for declarative hardware configuration and real-time trace debugging, replacing a **3 GB** Eclipse-based IDE with a single **10 MB** binary. Designed a Rust core that parses a human-readable `stm32.toml` config, validates pin and peripheral assignments against a hardware constraint database, and generates clean HAL initialization C code along with a ready-to-build CMake project. Implemented an LSP server that surfaces live pin-conflict diagnostics directly in the editor, and an ITM trace daemon that decodes CoreSight packets from OpenOCD and streams them over WebSocket to a real-time React dashboard, accessible through a lightweight VS Code extension.
Tech Stack: Rust, TypeScript, React, CMake, OpenOCD
- **SparkIDE** – *A beginner-friendly block-based IDE for Arduino (In Progress)*: Building a lightweight, visual IDE that brings Scratch-like block programming to Arduino development on Linux. Wrapped Google Blockly in a native PyQt6 desktop shell, loaded offline through QtWebEngine, so beginners can drag and drop blocks and watch the corresponding Arduino C++ generate live in a side panel. Authored **50+ custom blocks** covering the standard Arduino library (digital/analog I/O, timing, serial, variables, logic, and loops), and integrated `arduino-cli` for board discovery, one-click compile, and flashing, with build logs streamed into a docked output console.
Tech Stack: Python, PyQt6, JavaScript (Blockly), arduino-cli
- **Kissan-Connect** – *An Android-based marketplace app for farmers*: Built an Android marketplace app using Jetpack Compose to connect farmers directly with consumers. Implemented backend functionality using Django and Firebase to manage user authentication, product data, and listings, with Firebase handling auth and storage while the backend handled core application logic and communication with the Android client. Gained hands-on experience building backend-supported mobile applications with real user workflows.
Tech Stack: Python, Kotlin, Jetpack Compose, Firebase, Django

VOLUNTEERING EXPERIENCE

- ***Society for Promotion of Electronics Culture (SPEC)*** *NIT Hamirpur*
Volunteer – Executive *September '24 - Present*
 - **Electrothon 8.0:** Managed and coordinated the 8th iteration of an MLH-backed hackathon for **500+ students** across multiple colleges from scratch, overseeing technical and administrative logistics.
 - **Breakout Brigade 4.0:** Organised and directed a workshop for **120+ students** of NIT Hamirpur; as a result, Hardware was rated the best among the four workshops.
 - **Electrothon 7.0:** Organised an MLH-backed hackathon with **2500+ registrations** and administered venue arrangements for more than 500 students from nearby colleges.

HACKATHONS

- ***HackSecure - NITH:*** Built an Android cybersecurity application that let users quantify all data entering and leaving the device, developing a custom network tunnel and an on-device VPN to capture and inspect the traffic.
- ***Rekkathon - NITH:*** Built a portable network-security enhancer that protected against multiple cyber attacks (MITM, DDoS, phishing), blocked network-wide ads via private DNS routing, included fire and smoke sensing, and offered a live video monitoring feature for the admin.
- ***Hack 5.0 - NITH:*** Built an integrated Chrome extension that converted live sign language into readable English using per-frame hand recognition.

CERTIFICATIONS

- **HackerRank Python Developer Certification:** [Certificate](#) (February '25)
- **HackerRank SQL Developer Certification:** [Certificate](#) (January '25)